

University Interscholastic League
2024 – 2025 Junior High Number Sense Test B

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

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| <p>(1) $50 \div 0.25 =$ _____</p> <p>(2) $25 \times 0.12 =$ _____</p> <p>(3) $411 - 109 =$ _____</p> <p>(4) $981 \div 9 =$ _____</p> <p>(5) $\frac{11}{16} - \frac{1}{2} =$ _____</p> <p>(6) $2.25 - 0.25 =$ _____</p> <p>(7) $12 \times 14 =$ _____</p> <p>(8) $23^2 =$ _____</p> <p>(9) $12 + 8 \div 2 - 4 =$ _____</p> <p>*(10) $2025 \times 101 - 2024 =$ _____</p> <p>(11) $73 \times 77 =$ _____</p> <p>(12) The median of 11, 6, 20, 16 is _____</p> <p>(13) $132 \times \frac{9}{11} =$ _____</p> <p>(14) $26 \times 11 - 26 \times 6 =$ _____</p> <p>(15) $13 \times 10 \times 13 =$ _____</p> <p>(16) $461 + 28 + 32 =$ _____</p> <p>(17) $56^2 =$ _____</p> <p>(18) $31 + 25 + 19 + 13 + 7 =$ _____</p> <p>(19) MMXXIV = _____ (Arabic Numeral)</p> | <p>*(20) $454 \times 219 - 26 =$ _____</p> <p>(21) $95 \times 98 =$ _____</p> <p>(22) If $3x - 13 = 26$, then $x^2 =$ _____</p> <p>(23) $8\frac{3}{4} + 6\frac{11}{12} =$ _____</p> <p>(24) An $8\frac{1}{3}\%$ sales tax on \$48 is \$ _____</p> <p>(25) $4\frac{1}{2} \times 2\frac{1}{2} =$ _____ (mixed number)</p> <p>(26) $1.25 \times 2.4 \times 10 =$ _____</p> <p>(27) 4 gallons = _____ cubic inches</p> <p>(28) The GCD of 12, 24 and 16 is _____</p> <p>(29) How many whole numbers evenly divide 54? _____</p> <p>*(30) $109 \times 636 - 24 =$ _____</p> <p>(31) $27 \times 87 =$ _____</p> <p>(32) If a square, with area 9, and equilateral triangle have the same perimeter, the triangle side = _____</p> <p>(33) If $2.5 - 0.25 = n$, the $n^{-1} =$ _____</p> <p>(34) $-9^2 =$ _____</p> <p>(35) 3 pints = _____ ounces</p> <p>(36) 3 inches = _____ centimeters</p> <p>(37) $58\frac{1}{3}\%$ of 24 is _____</p> |
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- (38) $494 \times 111 =$ _____
- (39) The area of a square with a diagonal of 12 is _____
- *(40) 3 miles = _____ feet
- (41) For the sequence: 2, **n**, 10, **q**, 26, 37, . . . **q - n** equals _____
- (42) $4^4 - 4^3 =$ _____
- (43) $31 \times 75 =$ _____
- (44) The number of elements in $\{2, 3, 5, 6\} \cap \{1, 2, 3, 4\}$ is _____
- (45) $3 \text{ (base 4)} \times 32 \text{ (base 4)} =$ _____ (base 4)
- (46) The length of the edge of a cube with surface area 54 is _____
- (47) The perimeter of a right triangle with leg 9 and hypotenuse 15 is _____
- (48) The smallest prime number larger than 50 is _____
- (49) The volume of a rectangular solid that measures 6 by 8 by x is 96. What is x ? _____
- *(50) $66275 \div 25 =$ _____
- (51) What is the area of the trapezoid with bases $9\frac{2}{3}$, $7\frac{2}{3}$ and height 6? _____
- (52) What is the 8th term in the sequence: 1, 2, 3, . . . ? _____
- (53) $101 \times 935 =$ _____
- (54) Three times a number added to 5 is 38. The number is _____
- (55) What is the perimeter of a rectangle with diagonal 15 and length 12? _____
- (56) The area of a rhombus with diagonals 14 and x is 105. What is x ? _____
- (57) $125 \times 24 =$ _____
- (58) $(7^3 + 7 \times 6) \div 4$ has a remainder of _____
- (59) If $15 - 2x > 43$, then $x <$ _____
- *(60) $8311 \div 79 \times 101 =$ _____
- (61) $24 + (-24) \div 8 =$ _____
- (62) $24\frac{3}{7}$ weeks = _____ days
- (63) 15 meters/sec = _____ km/hr
- (64) $0.7333\dots =$ _____ (common fraction)
- (65) $\frac{3}{5} + \frac{5}{3} =$ _____
- (66) The volume of a circular cylinder with radius 8 and length 20 is $k\pi$. What is k ? = _____
- (67) The sum of the whole number divisors of 20 is _____
- (68) $\frac{5!+3!}{3!} =$ _____
- (69) 275% of 240 = _____
- *(70) 2 square yards = _____ inches
- (71) $286 \times 56 =$ _____
- (72) $58\frac{1}{3}\%$ = _____ (common fraction)
- (73) $30 \times 0.7333\dots =$ _____
- (74) The surface area of a sphere with a diameter of 6 is $k\pi$ and $k =$ _____
- (75) What is the probability of drawing an ace from a standard deck of 52 cards? _____
- (76) If the shortest distance between the points (6, 5) and (12, **x**) is 10, what is **x**? _____
- (77) $4^4 \div 16 =$ _____
- (78) 14% of **n** equals 9% of 28. **n** = _____
- (79) $21 + 19 + 17 + \dots + 1 =$ _____
- *(80) $6875 \times 321 =$ _____

2024 – 2025 University Interscholastic League Junior High Number Sense Test B – Key

(1) 200	*(20) 94430 – 104370	(38) 54834	(59) -14
(2) 3	(21) 9310	(39) 72	*(60) 10095 – 11156
(3) 302	(22) 169	*(40) 15048 – 16632	(61) 21
(4) 109	(23) $15\frac{2}{3}; \frac{47}{3}$	(41) 12	(62) 171
(5) $\frac{3}{16}; .1875$	(24) 4.00	(42) 192	(63) 54
(6) 2	(25) $11\frac{1}{4}$	(43) 2325	(64) $\frac{11}{15}$
(7) 168	(26) 30	(44) 2	(65) $2\frac{4}{15}; \frac{34}{15}$
(8) 529	(27) 924	(45) 222	(66) 1280
(9) 12	(28) 4	(46) 3	(67) 42
*(10) 192376 – 212626	(29) 8	(47) 36	(68) 21
(11) 5621	*(30) 65835 – 72765	(48) 53	(69) 660
(12) $13.5; 13\frac{1}{2}; \frac{27}{2}$	(31) 2349	*(50) 2519 – 2783	*(70) 2463 – 2721
(13) 108	(32) 4	(51) 52	(71) 16016
(14) 130	(33) $\frac{4}{9}$	(52) 8	(72) $\frac{7}{12}$
(15) 1690	(34) -81	(53) 94435	(73) 22
(16) 521	(35) 48	(54) 11	(74) 36
(17) 3136	(36) $7.62; 7\frac{31}{50}; \frac{381}{50}$	(55) 42	(75) $\frac{1}{13}$
(18) 95	(37) 14	(56) 15	(76) 13
(19) 2024		(57) 3000	(77) 16
		(58) 1	(78) 18
			(79) 121
			*(80) 2096532 – 2317218

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.